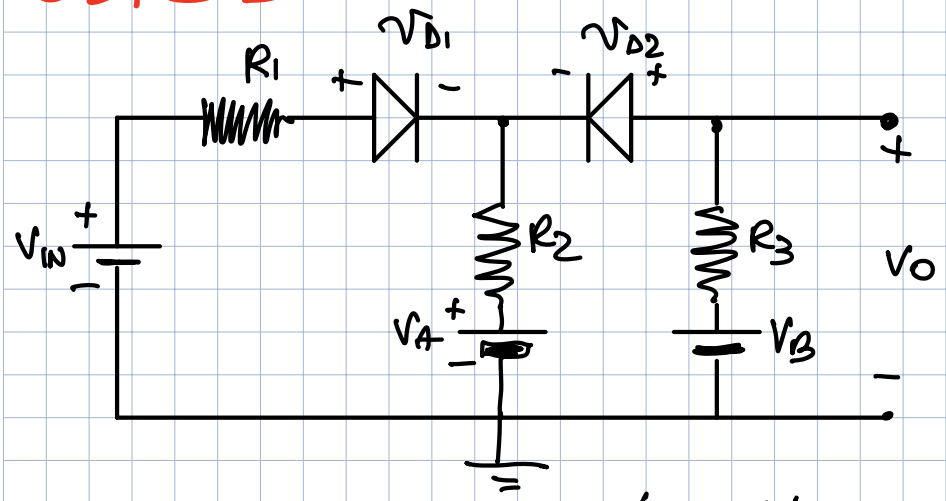
EDP2-3



$R_1 = 1{,}5\ k\Omega$
$R_2 = 10\ k\Omega$
$R_3 = 5\ k\Omega$

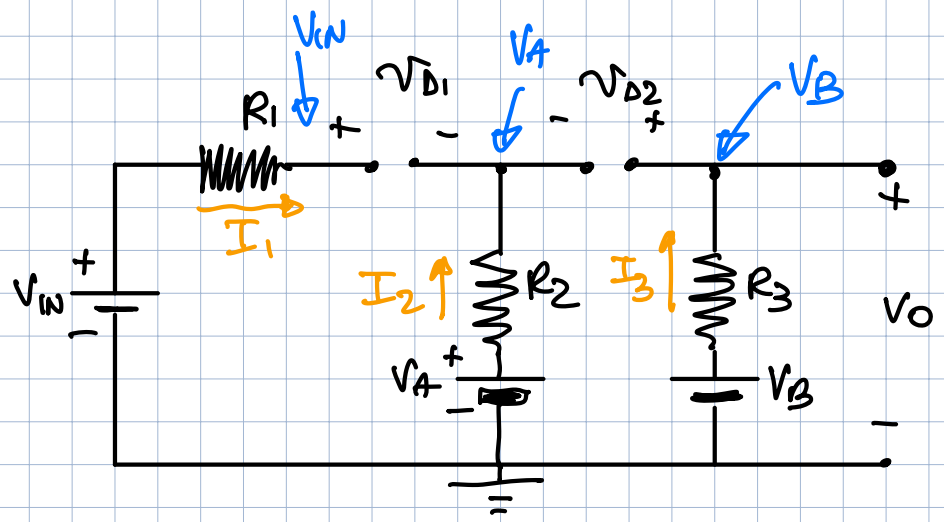
$V_A = 2{,}5V$
$V_B = 10V$

$V_{ON} = 0V$

TROVARE v_O QUANDO $V_{IN} = -5V$, $V_{IN} = 5V$, $V_{IN} = 10V$, $V_{IN} = 30V$

FACOLTATIVO! TROVARE $v_O = f(V_i)$

IPOTESI D1 = OFF, D2 = OFF



$I_1 = I_2 = I_3 = 0$

$v_{D1} = V_{IN} - V_A$

$v_{D2} = V_B - V_A$

a) $V_{IN} = -5V$

$\Rightarrow$ $v_{D1} = V_{IN} - V_A = -5V - 2{,}5V = -7{,}5V \Rightarrow$ D1 = OFF

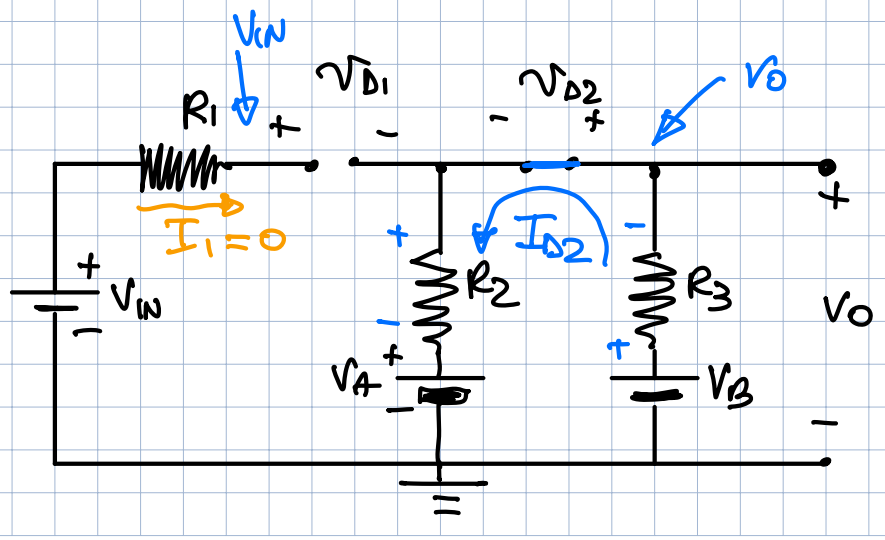
$v_{D2} = V_B - V_A = 10V - 2{,}5V = 7{,}5V \Rightarrow$ D2 = ON

HP D1 = OFF
D2 = ON

$I_{D2} = \dfrac{V_B - V_A}{R_2 + R_3} = 0{,}5\ mA > 0$

OK D2 = ON!

$v_{D1} = V_{IN} - V_O$



$V_O = V_B - R_3 I_{D1} = 10V - 5K\Omega \cdot 0{,}5mA = 7{,}5V$

OPPURE $V_O = V_A + R_2 I_2 = 2{,}5V + 10K\Omega \cdot 0{,}5mA = 7{,}5V$

OPPURE $V_O = V_A \cdot \frac{R_3}{R_2+R_3} + V_B \frac{R_2}{R_2+R_3} = \frac{5K\Omega}{15K\Omega} V_A + \frac{10K\Omega}{15K\Omega} V_B$

$= \frac{V_A}{3} + \frac{2}{3} V_B = \frac{2{,}5}{3} + \frac{20}{3} V = 7{,}5V$

$\Rightarrow v_{D1} = V_W - V_O = -5V - 7{,}5V = -5V < 0$

OK D1 = OFF

QUINDI

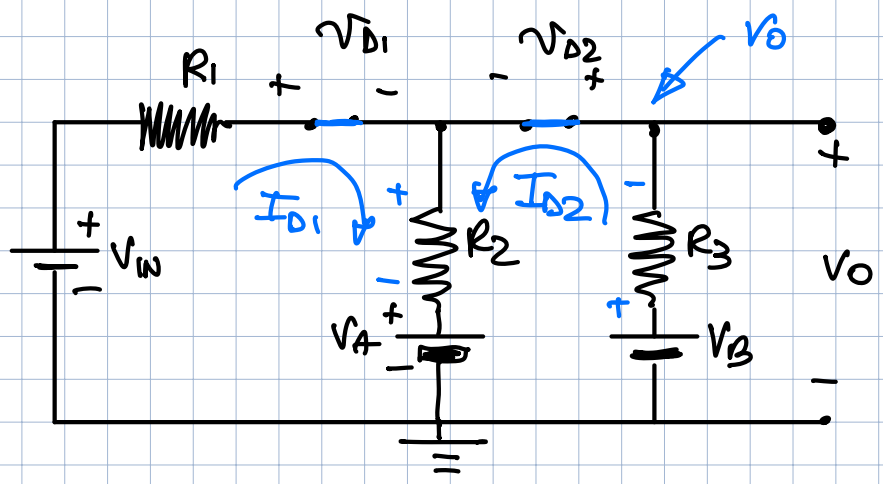
$$\boxed{\begin{array}{l} V_{IN} = -5V \\ D1 = OFF \\ D2 = ON \end{array} \quad e \quad V_O = 7{,}5V}$$

b) $V_{IN} = 5V$

$\Rightarrow$ $v_{D1} = V_W - V_A = 5V - 2{,}5V = 2{,}5V \Rightarrow D1 = ON$

$v_{D2} = V_B - V_A = 10V - 2{,}5V = 7{,}5V \Rightarrow D2 = ON$

HP D1 = ON
D2 = ON



SOVRAPP. EFF.

$V_O = V_{IN} \cdot \frac{R_2 // R_3}{R_1 + R_2 // R_3} +$

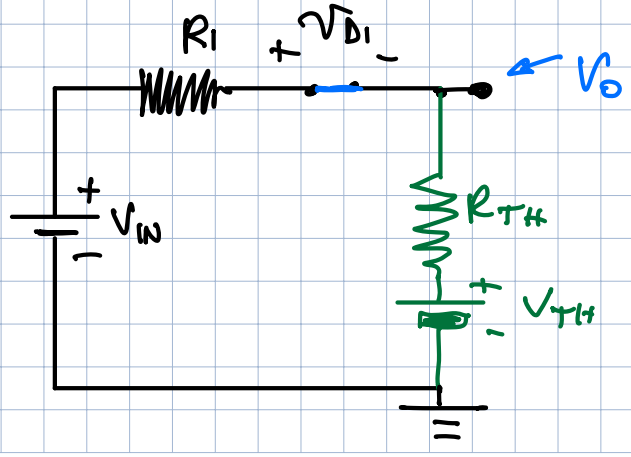
$+ V_A \frac{R_1 // R_3}{R_2 + R_1 // R_3} + V_B \frac{R_1 // R_2}{R_3 + R_1 // R_2} =$

$= V_{IN} \cdot 0{,}689 + V_A \cdot 0{,}103 + V_B \cdot 0{,}207 = 5{,}77V$

POTEVO SFRUTTARE IL CIRCUITO DI PRIMA, TROVARE EQ. THEVENINA DESTRA DI D1

$R_{TH} = R_2 // R_3 = 3{,}333K\Omega$

$V_{TH} = 7{,}5V$



$V_O = V_{TH} \cdot \frac{R_1}{R_1 + R_{TH}} + V_{IN} \frac{R_{TH}}{R_1 + R_{TH}} =$

$= 2{,}33V + 3{,}45V = 5{,}76V$

MA $I_{D1} = \frac{V_{IN} - V_O}{R_1} = -0{,}507\ mA < 0$ D1 NON È ON!

$I_{D2} = \frac{V_B - V_O}{R_3} = 0{,}848\ mA > 0$ OK D2 "ON"

⇒ IL CIRCUITO È ANCORA QUELLO DEL CASO a)

CON $V_{IN} = -5V$: D1 = OFF
D2 = ON

AVEVAMO TROVATO: $V_O = 7{,}5\ V$

$I_{D2} = 0{,}5\ mA > 0$ OK D2 = ON

$v_{D1} = V_{IN} - V_O = 5V - 7{,}5V = -2{,}5V < 0$

OK D1 = OFF!

SE STAVO ATTENTO/A, CONSIDERANDO IL CASO a) POTEVO CAPIRE DA SUBITO CHE ERA ANCORA D1 = OFF!

⇒ $V_{IN} = +5V$, D1 = OFF, D2 = ON e $V_O = 7{,}5V$

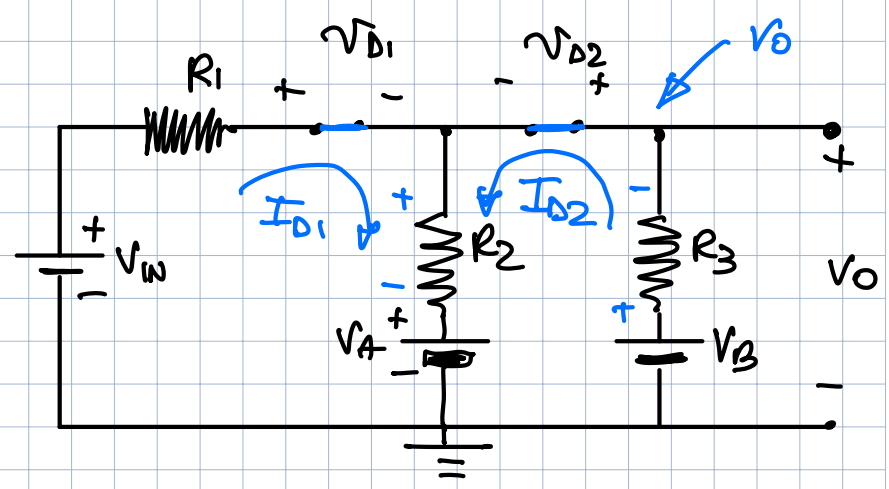
c) $V_{IN} = 10\ V$

$v_{D1} = V_{IN} - V_A = 10V - 2{,}5V = 7{,}5V$ ⇒ D1 = ON

$v_{D2} = V_B - V_A = 10V - 2{,}5V = 7{,}5V$ ⇒ D2 = ON

(ANCHE GUARDANDO I DUE CASI a) e b) CAPISCO CHE D1 È "ON", $v_{D1} = 10V - 7{,}5V = 2{,}5V > 0$!)

HP D1 = ON
D2 = ON



SOVRAPP. EFF.

$$V_O = V_{IN} \cdot \frac{R_2 // R_3}{R_1 + R_2 // R_3} +$$

$$+ V_A \frac{R_1 // R_3}{R_2 + R_1 // R_3} + V_B \frac{R_1 // R_2}{R_3 + R_1 // R_2} =$$

$= V_{IN} \cdot 0{,}683 + V_A \cdot 0{,}103 + V_B \cdot 0{,}207 = 9{,}22\,V$ (*)

$I_{D1} = \frac{V_{IN} - V_O}{R_1} = 0{,}52\,mA > 0$ OK D1 = ON

$I_{D2} = \frac{V_B - V_O}{R_3} = 0{,}156\,mA > 0$ OK D2 = ON

QUINDI

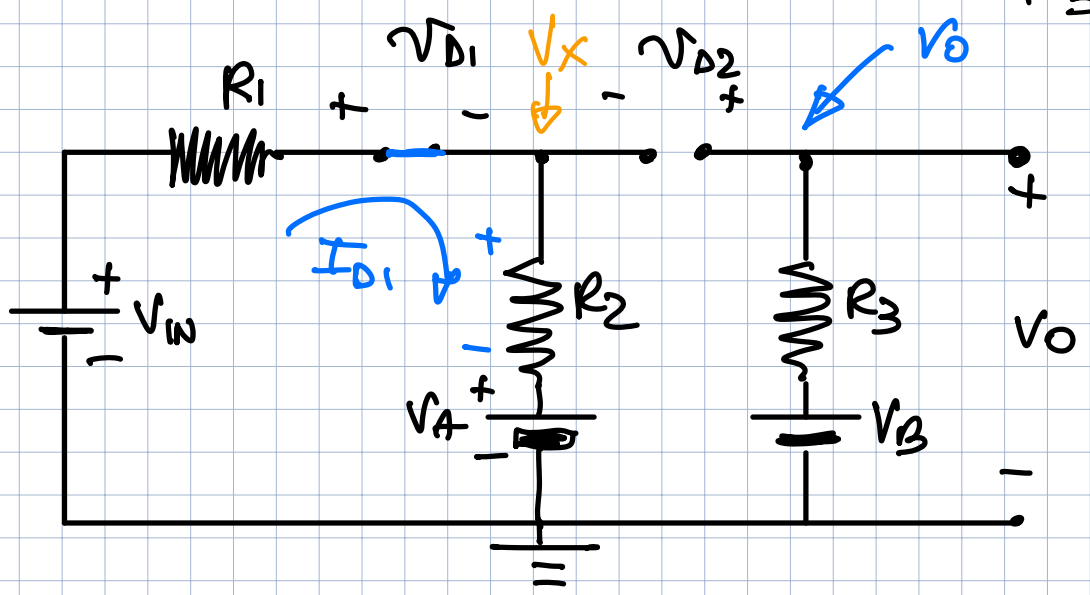
$V_{IN} = 10\,V$
D1 = ON e $V_O = 9{,}22\,V$
D2 = ON

d) $V_{IN} = 30\,V$

CON $V_{IN} = 30\,V$, SUPPONENDO I DUE DIODI "ON" COME IN c) TROVO CHE (DALLA RELAZ. PREC.)

* $V_O = V_{IN} \cdot 0{,}683 + V_A \cdot 0{,}103 + V_B \cdot 0{,}207$
$= 23\,V$

IN QUESTO CASO $I_{D2} = \frac{V_B - V_O}{R_3} = -2{,}6\,mA < 0$
$\Rightarrow$ D2 = OFF



$I_{D1} = \frac{V_{IN} - V_A}{R_1} = \frac{30\,V - 2{,}5\,V}{R_1 + R_2} = 2{,}39\,mA > 0$ OK D1 ON

$V_X = V_A + R_2 I_{D1} = 26{,}41\,V$

$$v_{D2} = V_B - V_X = -16{,}41 < 0 \quad \text{OK } D2 = \text{OFF}$$

$$\text{e } V_O = V_B \ !$$

QUINDI

$$\boxed{\begin{array}{l} V_{IN} = 10V \\ D1 = \text{ON} \quad \text{e } V_O = 10\,V \\ D2 = \text{OFF} \end{array}}$$

FACOLTATIVO – TRANSCARATTERISTICA

USANDO PARTE DEI RISULTATI PRECEDENTI

DA a) $\left.\begin{array}{l} D1 = \text{OFF} \\ D2 = \text{ON} \\ V_O = 7{,}5\,V \end{array}\right| \Rightarrow I_{D2} = \text{ON}$ e NON DIPENDE DA V_{IN}

$$v_{D1} = V_{IN} - V_O < 0 \Rightarrow V_{IN} < 7{,}5V$$

$$\Rightarrow \underline{\text{CON } V_{IN} < 7{,}5V} \Rightarrow \boxed{\begin{array}{l} D1 = \text{OFF} \\ D2 = \text{ON} \\ V_O = 7{,}5\,V \end{array}}$$

CON $V_{IN} > 7{,}5V \Rightarrow D1 = \text{ON} \rightarrow$ Vedi b) e c)

$$V_O = V_{IN} \cdot \frac{R_2 // R_3}{R_1 + R_2 // R_3} + V_A \frac{R_1 // R_3}{R_2 + R_1 // R_3} + V_B \frac{R_1 // R_2}{R_3 + R_1 // R_2} =$$

$$= V_{IN} \cdot 0{,}689 + V_A \cdot 0{,}103 + V_B \cdot 0{,}207$$

$$= V_{IN} \cdot 0{,}689 + 2{,}327\,V$$

$$I_{D1} = \frac{V_{IN} - V_O}{R_1} \Rightarrow \frac{0{,}311\, V_{IN} - 2{,}327V}{R_1} > 0$$

$\Rightarrow V_{IN} > 7{,}5\ V$ OK SEMPRE "ON"

$$I_{D2} = \frac{V_B - V_O}{R_3} = \frac{7{,}673 - 0{,}689 V_{IN}}{R_3} > 0$$

$$\boxed{V_{IN} < 11{,}136\ V}$$

RACCORDI: $V_O(7{,}5) = 7{,}5V$ OK

$V_O(11{,}13) = 10V$ DA VERIFICARE

QUINDI

CON $7{,}5V \leq V_{IN} < 11{,}13\ V \Rightarrow$ D1 = ON

D2 = ON

$V_O = 0{,}689 V_{IN} + 2{,}327V$

CON $V_{IN} > 11.13\ V$

CON $V_{IN} > 11{,}13\ V \Rightarrow$ D1 = ON

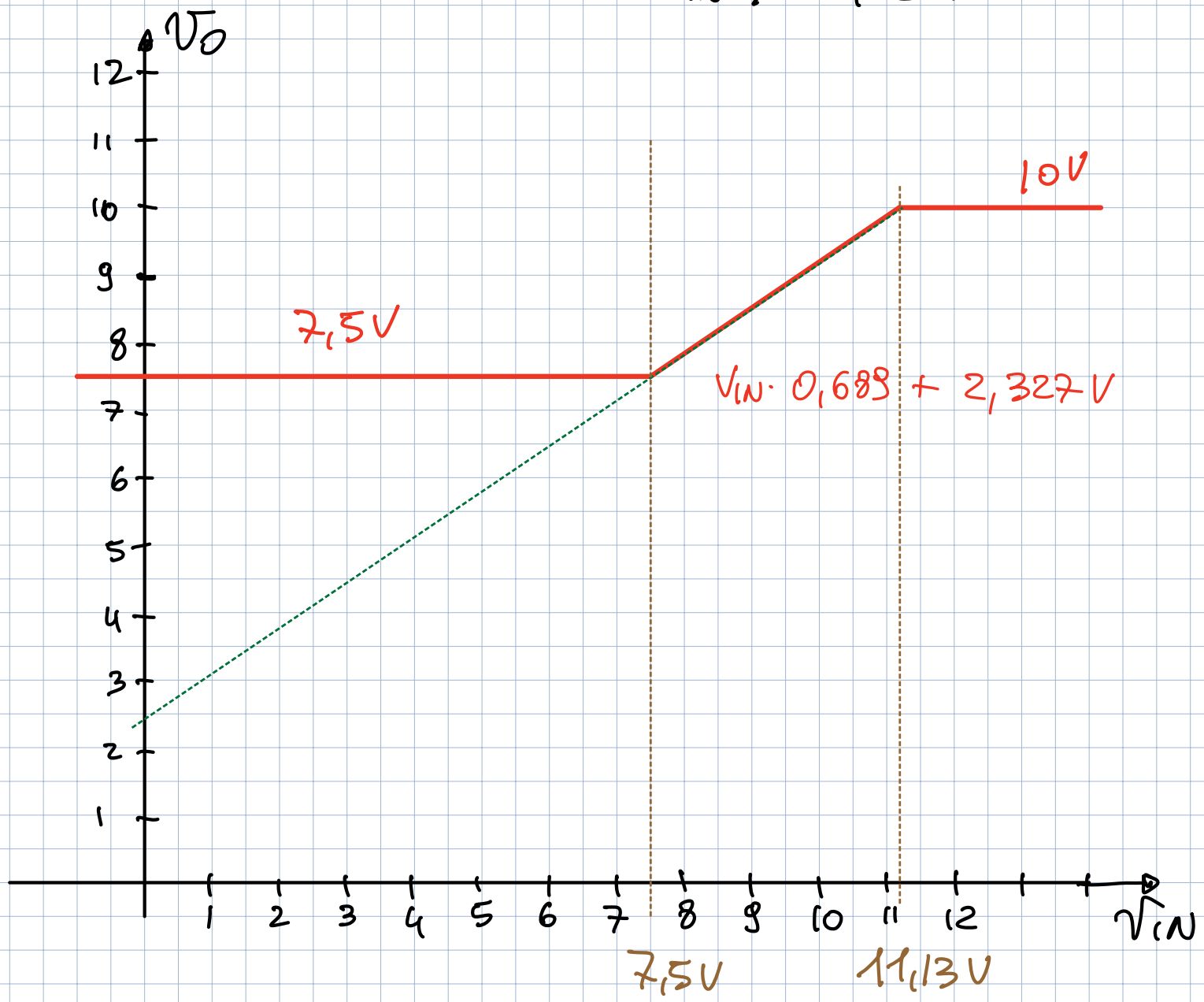
D2 = OFF $\rightarrow$ Vedi d)

$V_O = 10V$ (OK RACCORDO)

$$I_{D1} = \frac{V_{IN} - V_A}{R_1 + R_2} > 0 \qquad \underline{V_{IN} > 2{,}5V \quad OK!}$$

$$V_{D2} = V_O - V_X = 10V - (V_A + R_2 I_{D1}) < 0$$

$$10V < V_A + R_2 \left(\frac{V_{IN} - V_A}{R_1 + R_2} \right)$$

$$(10V - 2{,}5V)\frac{R_1+R_2}{R_1} < V_{IN} - V_A$$

$$V_{IN} > 11{,}13\,V \quad \text{OK}$$